Body Control Module - High-Level Design (Rev B)

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1. Component Overview

This document describes the High-Level Design of the Body Control Module software cluster, covering lighting, door, wind

Software Components

Component: BodyControlModuleComponent
Component: LightingManagerComponent
Component: DoorControlComponent
Component: MirrorControlComponent
The BodyControlModuleComponent acts as the central coordinator for all body domain functions.

2. Interfaces and Ports

Interface: LightingControlInterface
Interface: DoorStatusInterface
Interface: MirrorPositionInterface
PPort: LightingManager_PPort
RPort: DoorControl_RPort
PPort: MirrorControl_PPort
LightingManagerComponent provides LightingControlInterface via LightingManager_PPort. DoorControlComponent requir

3. Signals

Signal: HeadlampState_Signal
Signal: DoorAjar_Signal
Signal: IndicatorRequest_Signal
Signal: MirrorPosition_Signal

4. Dependencies

BodyControlModuleComponent depends on LightingManagerComponent for lighting state, on DoorControlComponent for

5. Functional Flow

On receiving an IndicatorRequest_Signal, LightingManagerComponent evaluates the current HeadlampState_Signal and i

6. Revision Notes

Added MirrorControlComponent and MirrorPositionInterface.
Tightened lamp actuation timing budget from 100 ms to 80 ms.